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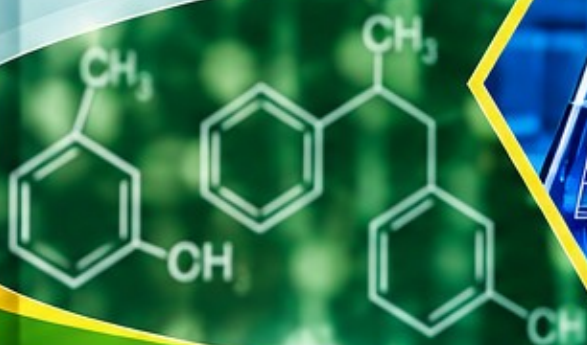
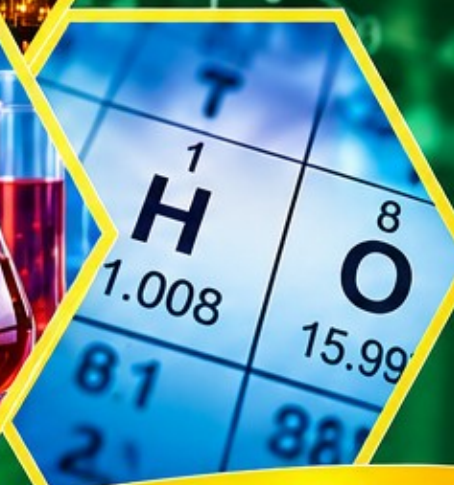
**PREPARE
PRACTICE
EXCEL**

Grade 12 CHEMISTRY

**REVISION, APPLICATION
AND EXAMINATION PRACTICE**

**BOOK
2**

- ✓ Very Brief Topical Notes
- ✓ Topical CBE Questions
- ✓ Mixed Concepts
- ✓ Grade 9 – 12 Quick Revision Q&A
- ✓ Five Final CBE Model Tests
- ✓ Complete Marking Schemes
- ✓ Quick Reference Section



CONCEPTS



APPLICATION



INVESTIGATION



EXAM SUCCESS

GRADE 12

SENIOR SCHOOL CHEMISTRY

BOOK 2

REVISION, APPLICATION & CBE EXAMINATION PRACTICE

Very Brief Notes • Application-Based Questions • Mixed Concepts • Grade 9–12 Q&A • 5 Model Tests

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Dedication

This book is dedicated to the loving memory of my late father, Simeon Nyatuka, whose guidance, sacrifices and belief in education continue to inspire my journey and the work I do as an educator.

To my mother, Hyrine Kerubo, for her love, strength and unwavering support throughout my life.

To my guardians, Peter Munde Mong'are and Teresa Kwamboka Sawe, for their guidance, encouragement and support.

To my friends, Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani, and many others, whose friendship and encouragement have been part of my journey.

To Her, my special one, for being a special part of my journey and for the place you hold in my heart. ❤️

To my fellow teachers, whose commitment to education continues to inspire the work of helping learners succeed.

And to every learner who opens this book: may your curiosity grow, your confidence strengthen and your effort lead you towards excellence.

To each of you, this book carries a part of your contribution to my journey.

— Hezekiah Omweri Nyatuka

Preface

Book 2 is designed for fast, focused revision. The notes are deliberately brief; the greater emphasis is on applying chemistry in real situations, interpreting evidence, solving calculations, investigating problems and justifying decisions.

The assessment tasks use CBE-style verbs such as investigate, interpret, analyse, predict, explain, apply, justify, evaluate, design and propose.

BOOK 2 RHYTHM: VERY BRIEF NOTES → APPLICATION → PRACTICE → EVIDENCE → REASONING → EXAMINATION

How to Use This Book

- Read the very brief notes and identify the key chemical idea.
- Master the equation, formula or diagram attached to the idea.
- Attempt the topical questions before checking the answer guidance.
- Use Mixed Concepts to connect topics rather than revising them in isolation.
- Use the Grade 9–12 Q&A section for rapid vertical revision.
- Attempt the five final model papers under timed conditions before checking the marking schemes.

CALCULATIONS: FORMULA → SUBSTITUTION → WORKING → ANSWER + UNIT

EXPLANATIONS: CLAIM → CHEMICAL EVIDENCE → REASONING

INVESTIGATIONS: VARIABLES → METHOD → OBSERVATION/DATA → INTERPRETATION → CONCLUSION → RELIABILITY/SAFETY

Contents

PART I - Grade 12 Topical Revision

1. Industrial Manufacture of Inorganic Chemicals
2. Carbon & Extraction of Metals
3. Energy Changes
4. Electrochemistry
5. Nuclear Chemistry
6. Alkanols
7. Alkanoic Acids
8. Detergents
9. Polymers

PART II - Mixed Concepts

PART III - Quick Revision Q&A: Grades 9, 10, 11 & 12

PART IV - Five Final CBE Examination Practice Tests

PART V - Marking Schemes

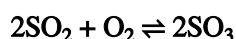
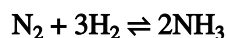
PART VI - Quick Reference & Exam Support

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PART I - GRADE 12 TOPICAL REVISION

1. Industrial Manufacture of Inorganic Chemicals

- Industrial manufacture aims at large-scale, economical production while balancing yield, rate, energy, safety, cost and environmental impact.
- Haber process: nitrogen + hydrogen $\rightleftharpoons$ ammonia; moderate temperature, high pressure and iron catalyst; unreacted gases are recycled.
- Contact process: sulfur dioxide is oxidised reversibly to sulfur trioxide using V_2O_5 at about 450°C .
- Solvay/Ostwald processes illustrate recovery/recycling of materials and controlled industrial conditions.



EXAM FOCUS: Use chemical evidence and reasoning, not unsupported statements.

Topical Application Questions

A fertiliser factory finds that higher pressure increases ammonia yield. Explain why. (2 marks)

Why is an extremely low temperature not used even though ammonia formation is exothermic? (2 marks)

State the role of the iron catalyst. (1 marks)

Why should unreacted gases be recycled? (2 marks)

Answer / Marking Guidance

Answer: 1. The product side has fewer gaseous molecules, so increased pressure favours ammonia formation.

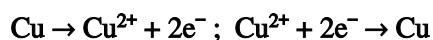
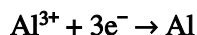
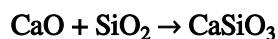
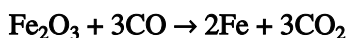
Answer: 2. The reaction rate would be too slow; industry uses a compromise between rate and equilibrium yield.

Answer: 3. It increases reaction rate by providing a lower-activation-energy pathway; it does not change equilibrium position.

Answer: 4. To reduce waste, improve raw-material utilisation and increase overall conversion.

2. Carbon & Extraction of Metals

- Diamond has a rigid 3-D covalent structure: very hard and non-conducting. Graphite has layers and delocalised electrons: soft and conducting.
- Metals above carbon generally require electrolysis; less reactive metals may be obtained by chemical reduction.
- Iron is extracted in a blast furnace; limestone removes silica as calcium silicate slag.
- Aluminium is extracted by electrolysis of molten aluminium oxide; cryolite lowers operating temperature and improves conductivity.
- Copper can be purified electrolytically. Corrosion of iron requires oxygen and water.



EXAM FOCUS: Use chemical evidence and reasoning, not unsupported statements.

Topical Application Questions

Why is aluminium extracted by electrolysis rather than carbon reduction? (2 marks)

Explain the role of limestone in the blast furnace. (2 marks)

Why is recycling aluminium particularly valuable? (2 marks)

State two conditions required for rusting. (2 marks)

Answer / Marking Guidance

Answer: 1. Aluminium is more reactive than carbon and its oxide is not reduced by carbon under normal extraction conditions.

Answer: 2. It forms CaO, which reacts with silica impurity to form calcium silicate slag.

Answer: 3. It reduces mining and requires much less energy than extracting aluminium from ore.

Answer: 4. Oxygen and water/moisture.

3. Energy Changes

- Exothermic processes release heat and have $\Delta H < 0$. Endothermic processes absorb heat and have $\Delta H > 0$.
- Calorimetry links temperature change to heat transfer. Always distinguish heat change of the surroundings from heat change of the reaction.
- Hess's law allows enthalpy changes to be obtained from alternative reaction pathways.
- Bond enthalpy method: energy required to break bonds minus energy released when bonds form.

$$q = mc\Delta T$$

$$\Delta H = H(\text{products}) - H(\text{reactants})$$

$$\Delta H = \Sigma E(\text{bonds broken}) - \Sigma E(\text{bonds formed})$$

$$q(\text{reaction}) = -q(\text{surroundings})$$

CAUTION: When water warms, $q_{\text{water}} > 0$ but $q_{\text{reaction}} < 0$. Use $q_{\text{reaction}} = -q_{\text{surroundings}}$.

Topical Application Questions

100 g of solution warms by 6.0°C . Calculate q if $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$. (3 marks)

Why is ΔH negative for an exothermic reaction? (2 marks)

State one reason an experimental enthalpy value may be lower in magnitude than expected. (1 marks)

What does a catalyst do to activation energy? (1 marks)

Answer / Marking Guidance

Answer: 1. $q = 100 \times 4.18 \times 6.0 = 2508 \text{ J} = 2.508 \text{ kJ}$.

Answer: 2. The reacting system loses heat to the surroundings.

Answer: 3. Heat loss to surroundings/apparatus or incomplete reaction.

Answer: 4. It provides an alternative pathway with lower activation energy.

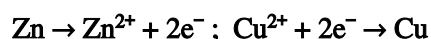
4. Electrochemistry

- Oxidation is loss of electrons; reduction is gain of electrons. Oxidation occurs at the anode and reduction at the cathode.
- In a galvanic cell electrons flow externally from anode to cathode. A salt bridge permits ion migration and prevents excessive charge build-up.
- Electrolysis uses electrical energy to drive chemical change. Faraday calculations link current, time and mass deposited.

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$$

$$Q = It$$

$$m = ItM/(zF), \quad F = 96500 \text{ C mol}^{-1}$$



REMEMBER: AN OX - RED CAT: oxidation at anode; reduction at cathode.

Topical Application Questions

Given $E^{\circ}(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$ and $E^{\circ}(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$, calculate E°_{cell} . (2 marks)

State the role of a salt bridge. (2 marks)

A current of 2.0 A flows for 30 min. Calculate charge. (2 marks)

Why should electroplating wastewater be treated? (2 marks)

Answer / Marking Guidance

Answer: 1. $0.34 - (-0.76) = +1.10 \text{ V}$.

Answer: 2. It permits ion migration, completes the internal circuit and prevents excessive charge build-up.

Answer: 3. $Q = 2.0 \times 1800 = 3600 \text{ C}$.

Answer: 4. It may contain metal ions that can contaminate water and should be recovered/removed before discharge.

5. Nuclear Chemistry

- Alpha radiation is strongly ionising but weakly penetrating. Beta has moderate penetration. Gamma is highly penetrating electromagnetic radiation.
- Half-life is the time for the number of undecayed nuclei, or activity, to fall to half its initial value.
- Fission splits heavy nuclei; fusion combines light nuclei. Radiation use requires control of time, distance and shielding.

$$N = N_0(1/2)^n$$

$$n = t/t_{1/2}$$

$$E = mc^2$$

EXAM FOCUS: Use chemical evidence and reasoning, not unsupported statements.

Topical Application Questions

A sample has half-life 6 h and initial mass 80 mg. Find mass after 18 h. (3 marks)

Why is an extremely long half-life often unsuitable for a short diagnostic procedure? (2 marks)

Distinguish fission from fusion. (2 marks)

State three radiation-protection principles. (3 marks)

Answer / Marking Guidance

Answer: 1. Three half-lives: $80 \rightarrow 40 \rightarrow 20 \rightarrow 10$ mg.

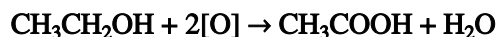
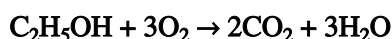
Answer: 2. The source remains radioactive for unnecessarily long, increasing handling/disposal burden and potential exposure.

Answer: 3. Fission splits a heavy nucleus; fusion combines light nuclei.

Answer: 4. Reduce exposure time, increase distance and use suitable shielding.

6. Alkanols

- Alkanols contain the hydroxyl group, -OH . General formula for saturated monohydric alkanols: $\text{C}_n\text{H}_{2n+1}\text{OH}$.
- Ethanol can be produced by fermentation of glucose. It burns completely in oxygen and can be oxidised to ethanoic acid.
- Industrial choices between production routes should consider raw materials, renewability, rate, purity, energy and cost.



EXAM FOCUS: Use chemical evidence and reasoning, not unsupported statements.

Topical Application Questions

State two suitable conditions for fermentation. (2 marks)

Why is combustion alone not a conclusive test for ethanol? (2 marks)

Give one benefit and one limitation of ethanol as a fuel. (2 marks)

What product forms when ethanol is oxidised fully under suitable conditions? (1 marks)

Answer / Marking Guidance

Answer: 1. Yeast and suitable warm conditions; limited oxygen is also appropriate.

Answer: 2. Other combustible organic substances may give similar combustion products; additional evidence is needed.

Answer: 3. Accept renewable feedstock/cleaner combustion as benefit; lower energy density/land-use or production constraints as limitation.

Answer: 4. Ethanoic acid.

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GRADE 12 CHEMISTRY - BOOK 2

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